

DeepSeek-V2: A Strong, Economical, and Efficient MoE Language Model

236B total parameters · 21B activated per token · 128K context length
[arXiv:2405.04434](https://arxiv.org/abs/2405.04434)

The LLM Scaling Dilemma: Performance vs. Cost & Speed

Training Cost

Larger dense models demand massive compute budgets. Scaling beyond 60B+ parameters becomes prohibitively expensive for most organizations.

KV Cache Bottleneck

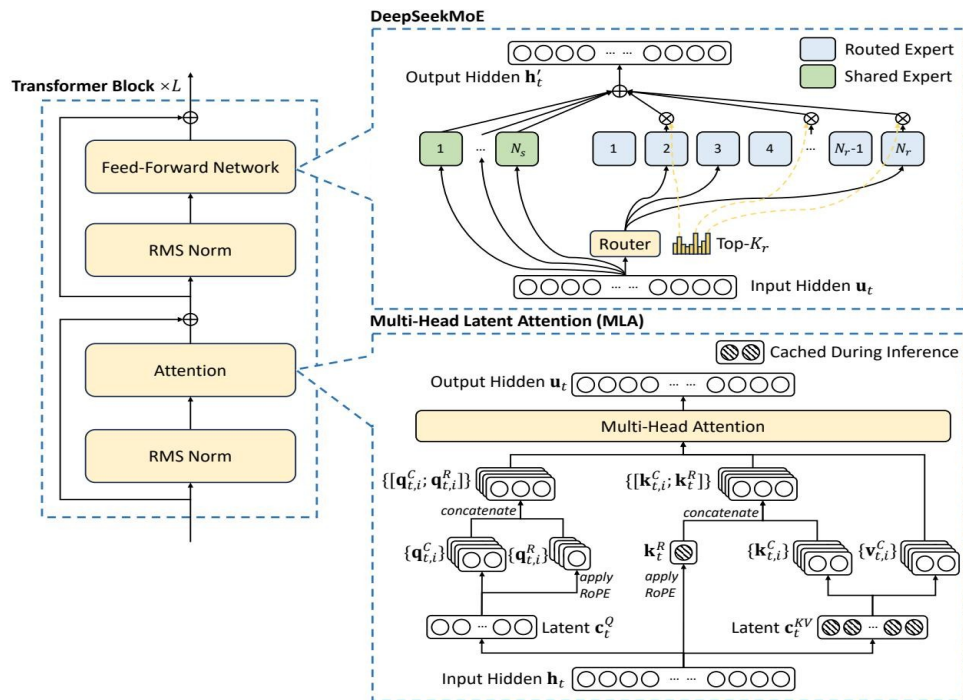
Standard Multi-Head Attention stores full key-value pairs per head per token, exploding memory usage during long-context generation.

Inference Throughput

Memory pressure and dense FFN computation limit serving capacity, making high-performance LLMs costly to deploy at scale.

Solution: Combine sparse MoE routing with compressed attention to break the cost-performance trade-off.

DeepSeek-V2 Architecture: MLA + DeepSeekMoE



Multi-Head Latent Attention (MLA)

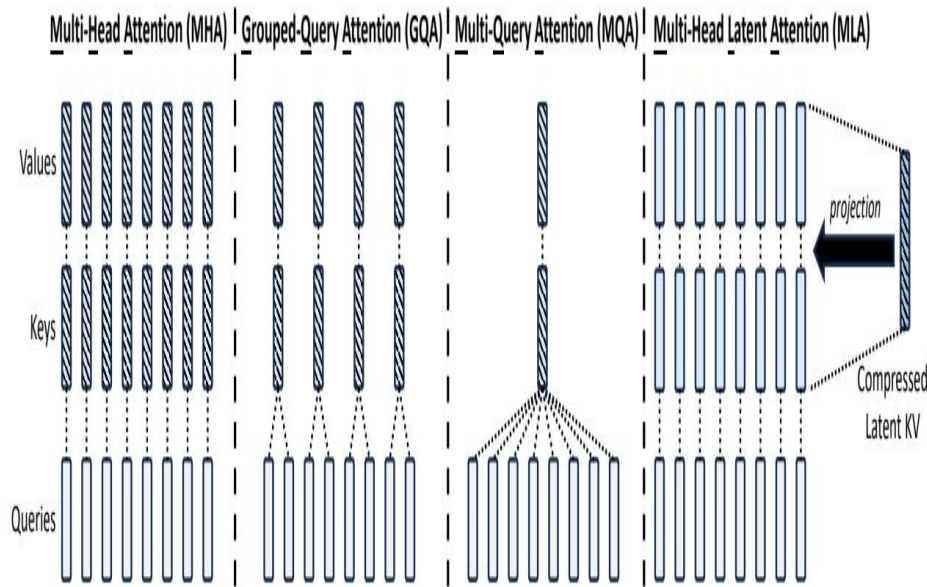
Low-rank KV joint compression + decoupled RoPE. Caches a compact latent vector instead of full K/V per head.

DeepSeekMoE Feed-Forward

Shared + routed experts with fine-grained segmentation, device-limited routing, and load-balancing losses.

236B total • 21B activated • 128K context

Multi-Head Latent Attention Compresses KV Cache by 93.3%



- Problem: MHA stores full K/V per head per token — cache grows linearly with layers, heads, and sequence length.
- GQA/MQA shrink cache but sacrifice expressiveness and accuracy.
- MLA innovation: compress keys and values into a single low-rank latent vector c_t^{KV} .
- During inference, reconstruct full K/V plus a decoupled RoPE positional key via learned projections.
- Result: better performance than MHA with 93.3% less KV cache.

MLA is a strictly better alternative to GQA/MQA for inference efficiency.

DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

Expert Design

- Shared experts (N_s): always active, capture common knowledge across all tokens.
- Routed experts (N_r): sparsely activated via top- K_r router per token.
- Fine-grained segmentation increases specialization potential without exploding parameter count.

Routing & Training Optimizations

- Device-limited routing: each token is routed to experts on at most M devices, cutting cross-node communication.
- Auxiliary losses enforce load balance across experts and devices.
- Token-dropping during training prevents expert overflow without affecting inference.

42.5% Training Cost Reduction and 5.76× Generation Throughput vs. DeepSeek 67B

42.5%

Training Cost Reduction

DeepSeek-V2 trains cheaper than the 67B dense predecessor despite 236B total params.

93.3%

KV Cache Reduction

MLA compresses the cache to a latent vector, slashing memory pressure during generation.

5.76×

Generation Throughput

Higher maximum throughput enabled by smaller cache and sparse activation.

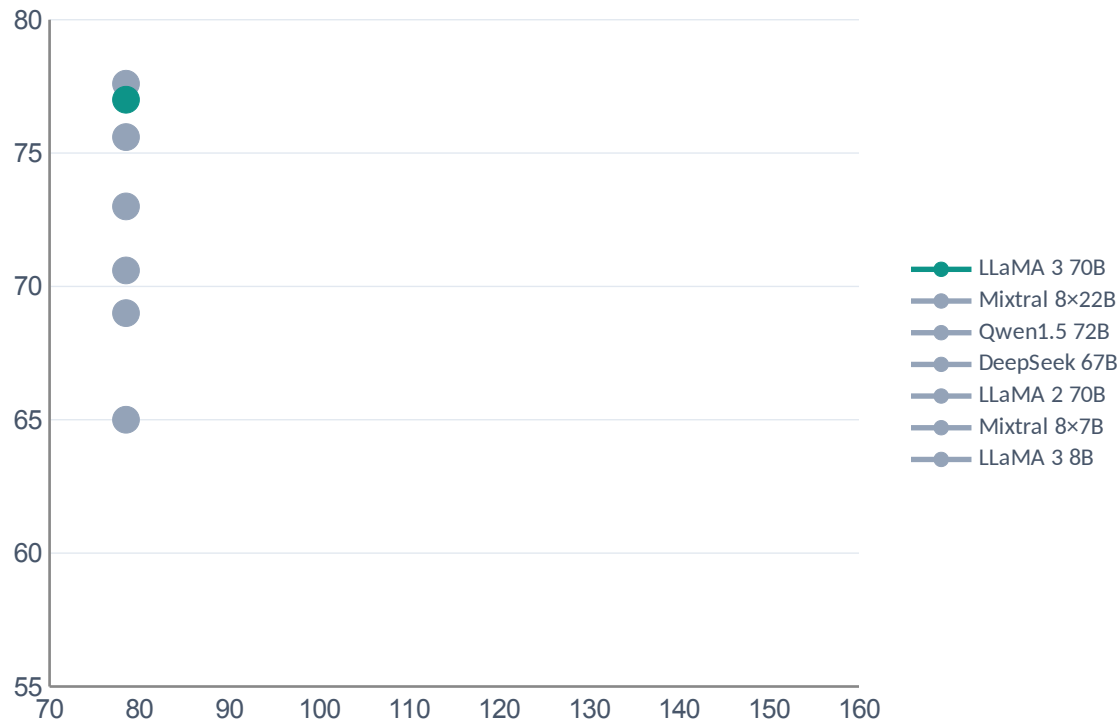
DeepSeek 67B (dense)

67B activated · Standard MHA · Dense FFN · Baseline throughput

DeepSeek-V2 (MoE + MLA)

21B activated · MLA (93.3% less cache) · Sparse MoE FFN · 5.76× throughput

DeepSeek-V2 Tops MMLU Among Open-Source Models at Only 21B Activated Parameters



Pareto-Optimal Position

- Highest MMLU score (78.5) among all listed open-source models.
- Activates only 21B parameters — 3.3× fewer than LLaMA 3 70B and 1.9× fewer than Mixtral 8×22B.
- Strictly dominates dense 70B-class competitors on the cost-accuracy frontier.

DeepSeek-V2 Chat (RL) Rivals Closed-Source Models on Chinese Benchmarks

Alignment Pipeline: SFT on 1.5M conversational sessions → GRPO-based reinforcement learning

38.9%

AlpacaEval 2.0

Length-controlled win rate

8.97

MT-Bench

Overall score

7.91

AlignBench

Overall (Chinese)

On AlignBench (Chinese), DeepSeek-V2 Chat (RL) outperforms all open-source models and most closed-source models.

Key Takeaways: Architectural Innovation Unlocks Efficient Scaling

MLA > MHA, GQA, MQA

Superior accuracy with 93.3% KV cache reduction via low-rank joint compression.

DeepSeekMoE Cuts Training Cost

42.5% cheaper than a 67B dense model by sparsely activating 21B of 236B parameters.

Pareto-Optimal MMLU

Highest open-source MMLU score (78.5) while activating far fewer parameters than 70B rivals.

5.76× Throughput Gain

Smaller cache + sparse activation make large-scale deployment practical and economical.

Strong Multilingual Alignment

Best open-source chat model in Chinese; competitive with closed-source systems in English.